

1. EXECUTIVE CASE OVERVIEW

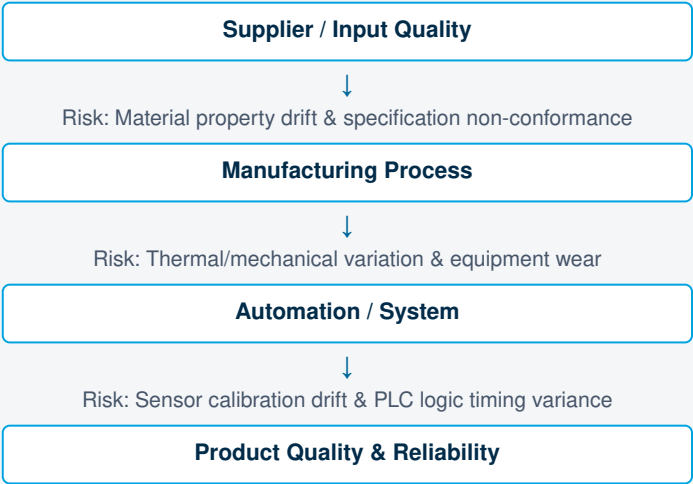
This document presents an illustrative industrial Quality Engineering scenario investigating interconnected manufacturing process variations, production defects, supplier input quality inconsistencies, and automation/system reliability. In modern industrial ecosystems, anomalies originating upstream in supplier materials, core manufacturing processes, or operational automation can propagate downstream, ultimately undermining final product consistency, performance, and field reliability. *Note: This is an independently developed illustrative technical case study and does not represent an actual Siemens internal project.*

2. PROBLEM STATEMENT

Industrial Production Challenge: An advanced industrial manufacturing system is experiencing recurring quality variation and component defects. The operational issue is non-isolated; multi-variable interactions between supplier input variations, equipment process instability, automation control behaviour, and latent product quality risks contribute to compounded downstream failures.

Core Objective: Systematically identify dominant root causes, reduce recurring defects and variance, strengthen prevention mechanisms, validate corrective actions, and establish sustainable statistical process control.

3. END-TO-END QUALITY CHAIN



4. ROOT-CAUSE ANALYSIS

Fishbone (Ishikawa) Analysis

Structured investigation framework categorizing potential failure modes across six dimensions:

Man | Machine | Method | Material | Measurement | Environment

5 Why Methodology

Iterative interrogative technique drilling past symptoms to system-level causes:

Observed Defect → Immediate Cause → Deeper Cause → System Cause → **Root Cause**

5. CAUSE PRIORITISATION (PARETO ANALYSIS)

Multiple potential contributors identified during investigation are evaluated using Pareto analysis. By ranking defect frequencies and financial impact, quality engineers separate vital few contributors from trivial many.

Purpose: Focus limited engineering resources on high-impact variables, prevent unfocused corrective actions, and ensure maximum reduction in overall process variance.

6. RISK ASSESSMENT (FMEA)

System FMEA and Process FMEA evaluate potential vulnerabilities at both architecture and operational levels:

RPN = Severity (S) × Occurrence (O) × Detection (D)

Technical Note: RPN acts as a prioritisation aid and is evaluated alongside engineering criticality and customer impact.

"The objective is not simply to identify many possible causes, but to systematically narrow the investigation toward the most significant risks and dominant contributors."

Industrial Quality & Process Reliability

7. DOMINANT CAUSE ISOLATION

Following Pareto and FMEA prioritisation, advanced statistical methodologies such as Shainin Red X are employed to isolate the single dominant variable governing the defect.

Multiple Variables → Candidate Screening → Comparative Tests
→ Red X Isolation → Targeted Action

9. PDCA IMPLEMENTATION

Systematic four-stage execution cycle ensuring disciplined problem resolution:

Plan	Define problem, root cause & objectives
Do	Execute improvement under controlled test
Check	Verify results against target metrics
Act	Standardize change & establish controls

8. PREVENTION & IMPROVEMENT

Poka-Yoke: Deploying error-proofing, physical interlocks, and automated verification to prevent human/process errors from escaping.

Kaizen: Driving continuous incremental improvements through practical shop-floor workspace optimizations.

10. PILOT & VALIDATION

Structured transition pathway ensuring robust industrial implementation:

Pilot (Controlled Scale) → Validation (Risk Verify) →
Standardisation (SOP Integration)

11. SPC & PROCESS CONTROL

Statistical Process Control monitors stable operations, distinguishing normal variation from special-cause anomalies requiring immediate corrective response.

12. COMPLETE QUALITY ENGINEERING METHODOLOGY FLOW

Stage / Tool	Primary Purpose	Stage / Tool	Primary Purpose
Fishbone	Explore possible causes	Shainin Red X	Isolate dominant cause
5 Why	Deepen investigation	Poka-Yoke	Prevent errors
Pareto	Prioritise contributors	Kaizen	Improve continuously
FMEA	Assess system/process risk	PDCA / Pilot	Implement & validate
Validation	Verify effectiveness	SPC & Control	Monitor & sustain improvement

13. POTENTIAL QUALITY OUTCOMES

- Reduced recurring defect rates & process variance
- Enhanced process stability & product consistency
- Stronger supplier quality assurance & controls
- Advanced automation reliability & defect prevention

14. KEY QUALITY ENGINEERING INSIGHT

"Quality should be built into the process through prevention, risk assessment, root-cause elimination, validation and statistical control — not dependent only on final inspection."

Disclaimer: This is an independently developed illustrative case study created by Jagriti Singh as part of the 100 Days TQM & Quality Engineering Challenge (Day 60/100). It is not an official Siemens project, internal Siemens case study, or representation of Siemens proprietary processes, systems, data, or results.